

Research Article

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Does the Strength of Incentives Matter for Elected Officials? A Look at Tax Collectors

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Abstract: In Pennsylvania local property taxes are collected by elected officials, known as tax collectors, whose compensation varies widely in both structure and level across municipalities. This paper analyses the existence of a pay-performance relationship for these officials. Using data on the percentage of property taxes that are actually collected at the municipal level, the paper finds that as the compensation tax collectors receive goes up, they collect more in taxes. This relationship is however true only for collectors who are compensated on a commission basis and not for collectors compensated on the basis of a flat salary.

Keywords: pay for performance, politician salary, tax collectors

1 Introduction

There is a growing theoretical and empirical literature that examines the effect of politician salary on a variety of measures such as performance of politicians while in office, the decision to run for election, and on the quality of politicians. This paper contributes to that literature by examining the institution of municipal tax collectors in the state of Pennsylvania. These tax collectors are elected officials who are responsible for the collection of property taxes for their respective municipalities. Using two different sources of data on tax collector compensation, this paper finds that an increase in their compensation is associated with an increase in the percentage of property taxes collected, but that this relationship holds only for collectors who are paid on the basis of a commission and not for collectors who are paid a flat salary.

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The predictions of the theoretical literature regarding the effect of politician compensation on the quality and performance of politicians are mixed. Work in the area of efficiency wages suggests that paying workers more reduces shirking because of the higher cost of being fired (Shapiro and Stiglitz 1984). These ideas are also central to the models in Barro (1973) and Ferejohn (1986) which predict that a higher salary by increasing the incumbent's payoff from being re-elected, likely spurs better performance on the job. On the other hand, a literature in personnel economics (Ariely, Bracha, and Meier 2009; Gneezy and Rustichini 2000) suggests that workers are often driven by intrinsic motivations and the impact of increasing external rewards might be to reduce the effect of intrinsic motivations via crowd-out. Thus given these ambiguous predictions, the analysis of the effects of compensation of elected officials on performance is an empirical matter.

Examining the relationship between politician compensation and performance however runs into a number of challenges. First, since politicians often play a role in deciding on their salaries, one cannot simply estimate the effect of compensation on politician performance without considering the possibility that causality runs in the opposite direction. For example, as documented by Di Tella and Fisman (2004) in the context of U.S. governors, politicians who demonstrate better on-the-job performance may be able to get the broader public to support pay raises for them. Recent papers attempt to overcome this problem either by exploiting large discontinuous changes in policy or by exploiting discontinuities of politicians' salaries with population size. Fisman et al. (2015) and Mocan and Altindag (2013) study politician performance in the European Union, exploiting a policy that equalized salaries of Members of the European Parliament (MEP) which had previously differed by as much as a factor of ten. These papers either find no effect of the increase in salaries on attendance and shirking (Fisman et al. 2015) or that it has a negative effect on attendance and the number of questions asked by the parliamentarians (Mocan and Altindag 2013).

Another empirical approach used to address concerns of reverse causality is to use a regression discontinuity research design. Exploiting a discontinuous change in salaries of municipal officials with a change in the population of Brazilian and Italian municipalities respectively, Ferraz and Finan (2011) and Gagliarducci and Nannicini (2013) find that more well-compensated politicians appear to have better performance in office. While Ferraz and Finan (2011) find that higher wages increase legislative productivity, resulting in more legislative bills and the provision of more public goods, Gagliarducci and Nannicini (2013) find that in municipalities with better-paid mayors, the speed of revenue collection (the ratio between revenue assessed and collected within the year) and the speed of payment (the ratio between outlays committed and paid within the year) is higher.

A third and final empirical approach used to address concerns of reverse causality is to conduct a randomized experiment as in Khan, Khwaja, and Olken (2016). Working with a provincial government in Pakistan, the authors were able to randomly allocate tax officials in the entire province's urban property tax department into one of three versions of performance-based pay schemes or a control group. Given the concerns around multitasking (Holmstrom and Milgrom 1991) and in particular about any negative impacts of high-powered incentives in the tax collection context, the three schemes used varied in the extent to which they based performance pay explicitly on taxpayer satisfaction and accuracy of assessment besides tax revenue, and the extent to which they allowed for subjective evaluation on the part of the tax department. Foreshadowing the results of this paper, the authors find that while instituting performance pay led to an increase in tax revenue (relative to the control group), the largest gains were obtained with a scheme which provided incentives purely based on revenue collected. Moreover, the authors also report finding little negative impact of the performance pay schemes on taxpayer satisfaction or in the accuracy of tax assessments.

The second empirical challenge in studying the relationship between pay and performance of politicians is in accurately measuring the performance of elected officials given that the goals of government are multidimensional and inherently hard to measure. Papers exploiting the reform that equalized salaries of MEPs focus on measures such as attendance (Fisman et al. 2015), the number of speeches, written declarations and reports drafted (Braendle 2015), and number of written and oral questions posed by the parliamentarians (Mocan and Altindag 2013). It is debatable whether these proxy measures truly capture the performance of MEPs who are enjoined to “act solely in the public interest and conduct their work with disinterest, integrity, openness, diligence, honesty, accountability and respect for the European Parliament's reputation.”¹

The relationship analyzed in this paper between pay and performance for municipal tax collectors within Pennsylvania potentially offers a number of advantages. First, I observe significant variation in both the structure and level of compensation of tax collectors across municipalities. While some municipalities compensate their collectors on the basis of a flat salary, others use a commission-based structure. For municipalities using a commission-based structure, there is large variation in the percentage of property taxes collected paid out as commission ranging from 1 to 5 percent and going as high as 10 percent for the smallest municipalities. Importantly, these variations are largely driven by variations in the codes that enable local government in Pennsylvania such as the Third Class

¹ Code of Conduct for MEPs available at: <http://www.europarl.europa.eu/meps/en/about-meps.html>.

City Code (1931), the Borough Code (1965), the First Class Township Code (1931), the Second Class Township Code (1933), and the Local Tax Collection Law (1945). Although there have been amendments over the years, the guidelines laid out in the original codes continue to influence the structure of tax collector compensation today. These variations in the compensation of tax collectors result in wide variation in the strength of incentives experienced by tax collectors. Few settings in the public-sector demonstrate such heterogeneity in the compensation structure of public officials and one that is not directly influenced by them.

Second, I am able to measure the performance of public officials in a manner that is possibly superior to prior work. For the tax collectors that I study, their key responsibility lies in collecting property taxes for their municipalities. I can estimate the performance of these tax collectors on the basis of what percentage of their municipality's predicted property taxes they were actually able to collect. Although this measure is far from perfect for reasons I describe in 4.1, a tax collector is in a position to influence the collection of property taxes – and hence this measure – more directly than a state's governor (or legislators) might be able to influence aggregate economic output of their state. Additionally because I am able to generate a measure of their performance, I can use it to comment on whether voters reward (or punish) a higher collection of property taxes by the tax collector at the ballot box.

The findings of the paper are as follows: higher compensation is associated with collection of a higher proportion of property taxes, but only for those tax collectors who are paid on a commission basis, with no relationship between pay and performance for tax collectors compensated on the basis of a flat salary. I rule out alternative explanations for the observed pay-performance relationship by virtue of the fact that the causal relationship does *not* appear to run from higher performance to higher compensation. I also conduct a falsification exercise which shows that higher commission rates have no relationship with the collection rate of the local *income* tax, suggesting that unobserved municipal differences do not drive the observed pay-performance relationship for municipalities that pay on a commission basis. Lastly, I do not find any statistically significant relationship between the percentage of taxes collected during the incumbent's term in office and either her decision to run for office again or her vote share in the next election cycle² suggesting that voters do not reward (or punish) higher collection of property taxes by tax collectors. These results perhaps help explain why higher

² The typical term of a tax collector is 4 years long. In some cases, I find terms that are 2 years long. Those are driven by resignations or retirements or deaths. Performance is measured for a 2-year term in such cases.

pay for salaried tax collectors does *not* induce a higher level of performance from them.

The paper is laid out in eight sections. Section 2 lays out the institutional setting while Section 3 develops the hypotheses and provides the outline of a theoretical model, leaving all formal elements to the Online Appendix. Section 4 describes the data sources and presents the econometric specification. Section 5 outlines the results with respect to the relationship between pay and performance including a number of robustness checks while Section 6 considers, and then rules out, alternative explanations for the positive pay-performance relationship. Section 7 examines the relationship between the performance in office and electoral outcomes with a discussion of the results and conclusions in Section 8.

2 Institutional Setting

Pennsylvania has a complex system of local government with over 2,500 municipalities – the second highest, next only to Illinois. Like local governments elsewhere, Pennsylvania’s municipalities rely on a mix of revenue sources of which the property tax and the local earned income tax are the two most important, each contributing 40–45 percent of all tax revenue.

Governance of a municipality varies based on the class of municipality and is laid out in the respective local government codes. Municipalities are classified as cities or boroughs or townships (which can be either of the first or of the second class). Beyond the governing body of a municipality (the city council or borough council or township board) each municipality also has a number of other independently elected offices, the office of tax collector being one such. As described in the Tax Collector’s Manual, “The local tax collector is the municipal officer designated to collect municipal and school real estate and personal taxes levied under the municipal codes, and in most cases, county real estate and personal taxes. In boroughs and second class townships, the office is designated as tax collector; in third class cities and first class townships, the elected treasurer is designated tax collector.”³ Tax collectors are elected for 4-year terms at municipal elections which are held in odd-numbered years.

The Tax Collector’s Manual also describes the minimal qualifications required for candidates. In third class cities, the treasurer must be a competent accountant, 21 years of age or more, and a resident of the city for a year before the election. An

³ The state of Pennsylvania has one city of the first class (Philadelphia), one of the second class (Pittsburgh), and one of the second class A (Scranton). The remaining 53 cities in the state are all cities of the third class.

individual does not have to be a CPA; s/he can be a qualified accountant through training and experience. In townships of the first class, the only qualification is being a registered voter of the municipality. In boroughs and townships of the second class, the tax collector must have resided in the municipality for one year before the election and continue to reside there during the term.⁴

As noted earlier, compensation of the tax collector is structured largely based on the local government codes, such as the Third Class City Code and the Borough Code. Within the broad parameters of these codes, the actual compensation of these officials is determined by the local government they collect taxes on behalf of. City treasurers are compensated in the form of an annual salary determined jointly by the city council, school board, and county commissioners. Compensation for tax collectors in townships and boroughs is set independently by the municipality, the school district, and the county. For townships of the first class, compensation is set by the township commissioners with the provision that annual compensation for township duties alone cannot exceed \$10,000.⁵ If the commissioners have not established a rate, the treasurer receives the statutory rate of 5 percent of taxes collected and 1 percent of other township funds received, subject to the \$10,000 maximum. Compensation in boroughs and townships of the second class can be in the form of a salary or a commission, with total compensation not to exceed 5 percent of the amount of taxes collected *except* for second class townships with populations less than 3000, where it can go up to 10 percent. A summary of the discussion in this section is provided in Table 1.

3 Hypotheses Development

In this section, I develop the hypotheses that focuses on whether the method of compensation (salary vs. commission) should matter for the effort exerted by the tax collector (and hence the percentage of taxes collected) and under what circumstances. It can help guide our thinking regarding what one should expect

⁴ The requirement that tax collectors seek the “Qualified Tax Collector” designation offered by Pennsylvania’s Department of Community and Economic Development (DCED) was in effect between October 2015 and December 2016 which falls outside the period analyzed in this paper.

⁵ Municipalities that enact home rule are allowed to exceed the \$10,000 cap for tax collectors and many do. Occasionally, non home-rule municipalities can exceed the cap as well. For example in 1991, the Board of Commissioners for Abington Township, Montgomery County approved the tax collector salary at \$10,000 for each of the next two years but provided \$4000 as compensation for collection of sewer rentals and another \$4000 for the collection of refuse fees, for a total compensation of \$18,000 (Ferry 1991).

Table 1: Summary of compensation scheme and titles used across municipalities of different classes.

Municipality type	Number of Municipalities	Median population (per 2010 Census)	Position title	Elected or appointed	Compensation scheme
Cities	56				
► First class	1	1,526,006	Revenue Commissioner	Appointed	Salary
► Second class	1	305,704	Treasurer	Appointed	Salary
► Second class A	1	76,089	Tax collector	Elected	Salary
► Third class	53	13,070	Treasurer	Elected	Salary
Townships	1547				
► First class	93	13,706	Treasurer	Elected	Mostly salary (capped at \$10,000)/commission (capped at 5 percent)
► Second class	1454	1978	Tax collector	Elected	Mostly commission (capped at 5 percent)/salary
Boroughs	958	1323	Tax collector	Elected	Mostly commission (capped at 5 percent)/salary
Total	2561	1914			

Notes: Author analysis based on successive issues of the Pennsylvania Manual and the Tax Collector's Manual. There are exceptions to the general principles outlined in the table. Notable among them is the fact that municipalities that have adopted home rule have often done away with the elected office of the tax collector and chosen to appoint someone to collect taxes. Second, while the commission rate for collecting property taxes in second class townships is capped at 5 percent, for townships with populations less than 3000, the commission rate may be as high as 10 percent of taxes collected.

when I regress the performance measure – the percentage of taxes collected – on various measures of compensation.

Four scenarios are constructed below. These scenarios differ along two dimensions. First, whether the re-election probability for an incumbent responds to the percentage of taxes collected varies across scenarios. Scenarios 1 and 2 consider the case when re-election probabilities do not depend on the percentage of taxes collected whereas scenarios 3 and 4 assume that re-election probabilities respond to the percentage of taxes collected during the previous term. Second, the method of compensation varies across scenarios. Scenarios 1 and 3 examine the case where the tax collector is compensated on the basis of a salary and scenarios 2 and 4 examine the case where the tax collector is compensated on the basis of a commission.

In the first scenario, characterized as S1, I consider a tax collector compensated on the basis of a salary whose re-election probability does not depend on the percentage of taxes collected. For such a tax collector, because salary and the re-election probability are assumed to be unresponsive to effort exerted, this leads to a somewhat obvious result that the optimal effort is 0.⁶

In the second scenario, S2, I consider a tax collector compensated on a commission basis whose re-election probability does not depend on the percentage of taxes collected. For such a tax collector, effort would be exerted to the point that the marginal benefit of effort (manifested in the form of a higher compensation) equals the marginal disutility of effort.

In scenarios 3 and 4, I introduce the possibility that the re-election probability may respond to effort exerted (and hence percentage of taxes collected). Although tax collectors compensated on a salary may not be motivated to exert higher effort as salaries go up if electoral incentives are non-existent, they may be so motivated in a world with electoral incentives. When the re-election probability depends on the percentage of taxes collected, it is reasonable to ask whether that probability goes up or down with an increase in taxes that are collected. Given that the median collection rate of property taxes is about 99%, it seems safe to assume that from the stand-point of the median homeowner, she would want the tax collector to faithfully collect all taxes that are due. Furthermore, renters would be unambiguously in favor of higher collection rates as they do not directly remit the property tax but benefit from spending on local public goods funded through

⁶ In practice, there may be a lower bound to effort as complete dereliction of duties may lead the taxing districts to take over these responsibilities. In such a case, effort exerted would equal the (exogenously set) lower bound.

property taxes.⁷ Consequently, I can expect that the median voter would want to see a higher collection of property taxes and re-election probabilities should be a weakly increasing (concave) function of the percentage of taxes collected.

Thus, in the third scenario, S3, I consider the case that the tax collector is compensated on the basis of a salary but her re-election probability depends on the percentage of taxes collected. In such a scenario, when re-election probabilities go up with effort, a salaried tax collector will exert positive effort because of electoral considerations even though her current-period compensation *does not* respond to effort.

Finally, in the fourth scenario, S4, the tax collector is compensated on a commission basis but as in S3, her re-election probability depends on the percentage of taxes collected. Here exerting effort results in a benefit both in the current period – through a higher compensation – but also in subsequent periods through an increase in the re-election probability.

The theoretical discussion is summarized in the form of Figure 1 which shows the predictions of how the optimal effort exerted changes depending on the presence or absence of electoral considerations and the compensation scheme used. In a nutshell, effort exerted goes up with an increase in the commission rate regardless of whether electoral considerations are present or absent whereas effort exerted goes up with an increase in salary only in the presence of electoral considerations.⁸

4 Data Sources and Empirical Specification

4.1 Data Sources

I utilize a variety of data sources for this study. There are two sources of data on compensation of tax collectors. The first source of data on compensation is the Pennsylvania Manual, a comprehensive guide to Pennsylvania's government.⁹ Each edition of the manual provides information on the compensation of tax

⁷ I ignore any second-order effects that result from part of the burden of the tax falling on renters.

⁸ I show in the Online Appendix that increasing compensation results in an increase in the optimal effort, when collectors are paid a salary and dynamic motives are present or whenever they are paid on a commission basis.

⁹ The Pennsylvania Manual serves as the “Blue book” (or the book of record) for the state of Pennsylvania. It has been published for more than 200 years by the state government itself and is described as a “complete compendium of state government information (Pennsylvania Manual, Vol. 113, page i).”

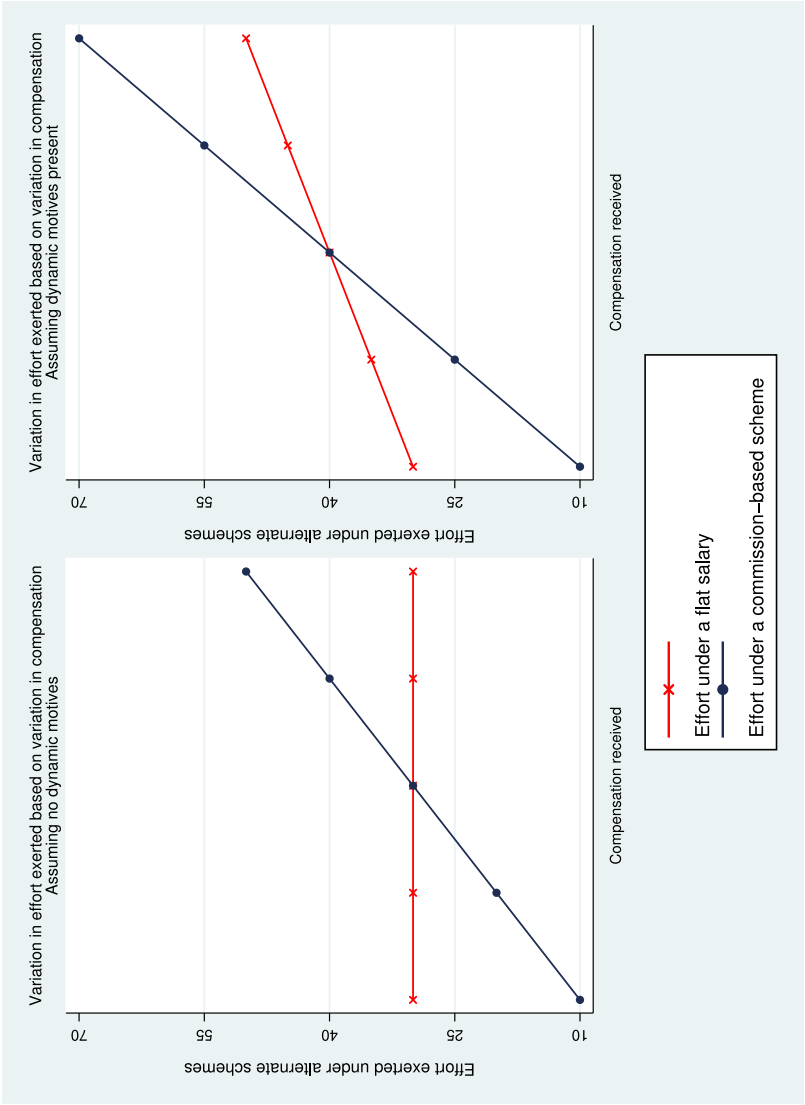


Figure 1: Summary of predictions of theoretical model – variation in effort in response to variation in compensation, depending on presence (or absence) of electoral considerations.

collectors from approximately 130 of the largest municipalities within Pennsylvania. The second comes from a 2004 survey conducted by the Pennsylvania Economy League (PEL)¹⁰ of municipalities in nine counties in central and eastern Pennsylvania regarding their tax collection practices (Table 2).¹¹

■ Q3

There are a number of advantages of the data included in the 2004 survey by the PEL. First, it includes compensation data on a larger number of municipalities compared to the number covered by the Pennsylvania Manual. Second, because it includes data on a number of smaller municipalities (generally boroughs and townships of the second class), the municipalities covered by the PEL survey are more representative of the state as a whole. For example, across a variety of dimensions such as size, population density, and percent of housing units that are owner-occupied, the municipalities included in the PEL sample are similar to the overall state.¹² One can also infer the representativeness of the PEL survey based on the fact that about 68 percent of the 300 municipalities included compensate their collectors on a commission basis – a number that is close to the 63 percent estimate provided in a report by the Legislative Budget and Finance Committee.¹³ Moreover, there is considerable variation in the percentage of property taxes paid to the tax collector as compensation in the sample.

The criteria for inclusion in the Pennsylvania Manual makes the data from this source less representative of the state as a whole. As the Manual provides data on compensation of local officials only for cities and for townships and boroughs that have more than 10,000 people, depending on the vintage of the Manual, I have data for fewer than 130 of the roughly 2500 municipalities in the state.

10 The Pennsylvania Economy League (PEL) is a non-partisan entity in existence since 1936. It emphasizes comparative research and policy analysis, and over time has obtained a “reputation for objectivity, integrity, and consensus building.” <https://digital.library.pitt.edu/islandora/object/pitt%3AUS-PPiU-ais197806/viewer>.

11 Table 2 summarizes the context specific to each county for which data were provided. The response rate to the survey was 74%. There is no evidence that non-response is systematically correlated with the municipal millage rate or the market value of real estate per capita and therefore any bias induced by non-response does not seem to be of first-order importance.

12 The average population for municipalities included in the PEL survey is about 4100 and although that is less than the average population of ~4900 for municipalities not in the PEL survey, that observation is entirely driven by the two largest cities – Philadelphia and Pittsburgh – not being included in the PEL survey. Taking those two cities out, the average population for municipalities not included in the PEL survey is about 4000 – very similar to the average for municipalities in the PEL sample. On population density as well, the PEL sample looks representative of the state as a whole, with both sub-samples averaging about 1200 people/sq. mile. The percent of housing units that are owner-occupied is also similar with about 83 percent units owner-occupied in the PEL sample and municipalities that are not in the PEL survey.

13 <http://lbfc.legis.state.pa.us/Resources/Documents/Reports/400.pdf> (p. S-3).

Table 2: Context specific to each county for which data were provided in the PEL survey.

County	Number of boroughs and townships	Municipalities with no property tax	Non- response	Special cases that cannot be used	Insufficient/ incomplete responses	Usable responses
Berks	73	2	10	0	3	58
Blair	23	0	8	1	0	14
Dauphin	39	2	10	1	2	24
Lackawanna	38	2	11	0	1	24
Lehigh	23	0	2	1	0	20
Luzerne	72	2	20	3	1	46
Lycoming	51	0	9	1	2	39
Northampton	36	0	9	3	0	24
York	71	1	17	2	0	51
Total	426	9	96	11	10	300

However, in this sample, tax collectors are more likely to be paid a flat salary which makes it possible to analyze and comment on the effects of a salary-based compensation scheme with greater confidence than would be possible using the PEL survey alone. Second, given the larger size of the municipalities covered by the Manual, nearly half of the state's 12 million people live within the borders of municipalities included in this sample. Last but not least, another advantage of the Pennsylvania Manual data is that they are available over time making it possible for me to construct a panel and introduce municipal fixed effects in our estimation. Thus these two data sources end up complementing each other and the fact that my conclusions remain robust across the datasets is reassuring.

In order to obtain a sense of the efficiency of the property tax collection system at the municipal level, I turn to data on municipal finances available from the Pennsylvania Department of Community and Economic Development (DCED). The data most relevant to us pertain to the assessed value of real estate, the millage rate chosen by the municipality, and property taxes that were actually collected. I can use these data to define a measure of performance, the percentage of property taxes collected as:

Percentage of property taxes collected

$$= \frac{\text{Property taxes collected}}{(\text{Millage rate}/1000 \times \text{assessed value of real estate})} \times 100$$

For example, consider Penn Township in York County. For 2009, their tax rate on the assessed value of a property was 2.3 mills and the total assessed value of real estate (not including tax-exempt property) was \$1,073,872,660. Thus the property taxes that should have been collected for the year equal \$2,469,907. However, \$2,440,026 was actually collected, suggesting that 98.8 percent of the taxes due were actually collected. The percentage of property taxes collected, averaged over the term of a tax collector, is the dependent variable included in the analysis when examining the existence of a relationship between pay and performance. It is worth pointing out that the millage is decided by the governing body of the municipality (such as a city council) while a board appointed by the county ("Board of Property Assessment, Appeals and Review") decides on the assessed value of real estate, *independently from the tax collector*.¹⁴ The tax collector however is in charge of mailing tax bills, collecting payment, following up on a routine basis with individuals or mortgage processors to ensure timely

¹⁴ There is an important literature on political incentives for tax assessors (e.g. Makowsky and Sanders 2013; Ross 2011). In Pennsylvania, collection and assessment are separate and this paper focuses on the collection.

collection. As a result, she plays a unique role in influencing the percent of property taxes that are being actually collected by the municipality.¹⁵

Although I view my ability to construct a measure of an elected official's performance as a key strength of this paper, I would be remiss if I did not note that this measure – percentage of property taxes collected – is noisy and imperfect for a few reasons. First, Pennsylvania voters through a Constitutional amendment passed in 1997 authorized local taxing authorities to exclude from taxation some portion of the value of primary homes (the homestead exemption) and farmland (the farmstead exclusion). Unfortunately I lack data on whether a municipality has chosen to adopt the homestead and farmstead exclusions and if so, what proportion of properties (and what proportion of the assessed real estate value) within their municipal borders qualifies. Second, Pennsylvania, like other states, can initiate a reassessment of a property if there are substantial improvements made to it such as the addition of a new deck or a pool. It is unclear when during the year those reassessments are done and how quickly such updates to property are reflected in the data on assessed values available from the Pennsylvania Department of Community and Economic Development. A similar argument applies for changes to the tax rate (millage) that occur within the course of a given year. It is unclear whether the tax rates available to us correspond to the rate in effect at the beginning of the year or at the end of the year or some weighted average thereof. On account of these limitations, the measure of the percentage of property taxes due that are actually collected is not perfect and it likely suffers from some mismeasurement.¹⁶

The last source of data I use pertain to elections for tax collectors. These are available from the websites of the Boards of Elections for the counties in which the municipalities are located. Election data prior to the early to mid-2000s are typically not available in electronic form. I pick 2003 as the starting point and look for which of Pennsylvania's 67 counties have data on elections available electronically with the County Boards of Elections. Ultimately I am able to obtain

15 Municipal tax collectors are not allowed to exercise discretion in establishing repayment plans for delinquent taxpayers to avert foreclosure or provide forbearance in light of challenging economic circumstances. The recent economic crisis stemming from COVID-19 provides a number of examples of where individual municipalities or school districts have chosen to exercise discretion to extend due dates but it is the governing bodies of those entities that have made those decisions rather than the tax collectors themselves.

16 A thorough examination suggests that the measurement error in the dependent variable is classical in nature. Thus, while this "measurement error results in a larger error variance than when the dependent variable is not measured with error" (Wooldridge 2002, p. 72), it does not induce a bias in the estimates. Appendix Table A.1 offers some evidence that is consistent with this claim.

data from 16 counties that have election data for the period between 2003 and 2009. These data are then subsequently matched with the data on tax collector performance to examine if performance in office has any relationship with the likelihood of an incumbent tax collector being re-elected and other proxies of electoral incentives.

Summary statistics are presented in Table 3.

Table 3 clearly indicates the wide variation in the compensation of tax collectors across municipalities. Additionally, one can discern wide differences in compensation based on the two sources – the 2004 PEL survey and the Pennsylvania Manual – reflecting primarily differences in size of the municipalities covered by these sources. The largest municipalities which appear in the Pennsylvania Manual are more likely to pay their tax collectors higher salaries than the smaller municipalities, which appear in the PEL survey. The largest municipalities are also more likely to pay on the basis of a flat salary than on a commission basis.

Table 3 also suggests that the absolute amount provided as compensation to these tax collectors is modest and it raises the question of whether these are full-time or part-time positions. In general, for the smaller municipalities, the tax collector position is part-time whereas it tends to be a full-time position for the larger municipalities. Moreover, it is important to note that the tax collector generally receives compensation from three taxing districts – the county, the school district, and the municipality – and Table 3 only captures that portion of the compensation provided by the municipality.¹⁷ Finally, about two-thirds of tax collectors also collect other taxes (such as occupational taxes) or fees on behalf of their local taxing districts and are compensated for doing the same (Legislative Budget and Finance Committee 2011). Thus, the overall compensation

¹⁷ Although the lack of data on compensation offered by the county and school district is a limitation of the study, I do not believe it induces any significant bias in the estimates, largely because taxing districts such as counties and municipalities tend to make decisions independently as to how to compensate their tax collectors. To the extent that any systematic evidence exists on compensation arrangements chosen by these taxing entities, a report prepared by the Pennsylvania Legislative Budget and Finance Committee notes that while a majority of municipalities (63 percent) compensate tax collectors on the basis of a percent of revenue collected, a significant majority of counties (72 percent) and school districts (85 percent) compensate local tax collectors on a per bill basis. <http://lbfc.legis.state.pa.us/Resources/Documents/Reports/400.pdf> (p. 28) One concrete example that illustrates the diversity of collection methods used is the York Suburban School District in York County which encompasses two townships – Spring Garden and Springettsbury. Although each tax collector is paid an average of \$3.55 for each school district real estate bill prepared and sent, I observe that while Spring Garden pays its collector a flat salary, neighboring Springettsbury pays its collector on a commission basis.

Table 3: Summary statistics.

	<i>N</i>	Mean	Standard deviation	<i>p25</i>	<i>p75</i>
Compensation data from the Pennsylvania Manual (data from 1985, 1989, 1993, 1997, 2001, and 2005)					
Municipalities compensating solely on a commission basis: Amount paid (averaged across all years)					
In absolute terms	34	\$15,186	\$13,017	\$7,111	\$20,310
As a percentage of property taxes collected	34	2.83	1.25	2.00	3.50
Municipalities compensating solely on a salary basis: amount paid					
From the 1985 Manual	108	\$9,645	\$7,792	\$4,650	\$12,500
From the 1989 Manual	110	\$11,161	\$9,563	\$5,000	\$15,000
From the 1993 Manual	126	\$14,782	\$11,553	\$6,000	\$20,400
From the 1997 Manual	108	\$17,729	\$13,601	\$8,398	\$23,750
From the 2001 Manual	111	\$18,864	\$14,805	\$9,400	\$26,000
From the 2005 Manual	119	\$19,423	\$17,980	\$9,800	\$28,000
As a percentage of property taxes collected (average)	680	0.96	1.35	0.31	1.17
Compensation data from the 2004 Pennsylvania Economy League (PEL) survey					
Municipalities compensating solely on a commission basis: amount paid					
In absolute terms	181	\$4,736	\$5,589	\$1,497	\$6,466
As a percentage of property taxes collected	181	4.29	1.57	3.13	5.00
Municipalities compensating solely on a salary basis: amount paid					
In absolute terms	47	\$6,048	\$5,066	\$2,500	\$8,795
As a percentage of property taxes collected	47	1.22	0.867	0.49	1.72

Table 3: (continued)

	<i>N</i>	Mean	Standard deviation	<i>p</i> ₂₅	<i>p</i> ₇₅
Election data from the County boards of elections					
Probability incumbent runs again for office	731	0.743	0.437	0	1
Probability incumbent wins office if she runs again	543	0.996	0.061	1	1
Incumbent's vote share	543	96.7	9.71	99.2	100
Incumbent's vote share (when faced with an opponent)	42	66.7	9.61	61.2	74.1
Percentage of taxes collected					
Sample matched with data from Pennsylvania Manual	716	99.4	2.65	98.0	100.5
Sample matched with the PEL survey	228	100.0	2.06	98.7	101.2
Sample matched with the election data	716	99.6	6.65	98.3	100.7

Notes: Data on compensation from the Pennsylvania Manual is drawn from six issues: volumes 107, 109, 111, 113, 115, and 117 and they provide tax collector compensation for the years 1985, 1989, 1993, 1997, 2001, and 2005 respectively. Data on compensation of municipal tax collectors from the Pennsylvania Economy League (PEL) is for 2004 and pertains to nine counties in Central and Eastern Pennsylvania (Berks, Blair, Dauphin, Lackawanna, Lehigh, Luzerne, Lycoming, Northampton, and York). Election data are for the period between 2003 and 2009 for 16 counties (Berks, Butler, Centre, Fulton, Lancaster, Lehigh, Luzerne, Mercer, Monroe, Montgomery, Northampton, Pike, Schuylkill, Venango, Washington, and Westmoreland) and are from the respective County Boards of Elections. Data on percentage of predicted property taxes actually collected are based on author's calculations using the municipal financial statistics compiled by the Pennsylvania Department of Community and Economic Development (DCED). Property taxes for each municipality are predicted as the product of their assessed value of real estate and the municipal millage rate. Percentage of taxes collected is the ratio of property taxes actually collected to predicted property taxes multiplied by 100.

of a tax collector is significantly higher than a casual inspection of Table 3 might suggest.

Table 3 also indicates that collection rate of property taxes is very high, reflecting the high compliance associated with the property tax¹⁸ and in fact, I observe collection rates that are occasionally higher than 100 percent. This is partly explained by the fact that in some instances, a large owner of property who was delinquent in a particular year may pay in a subsequent year causing the collection rate for that year to go above 100 percent. Additionally, this is affected by how and when penalties get assessed. Bills paid prior to April 30 of each year qualify for a two percent discount whereas a 10 percent penalty is assessed for payments made after July 1. Therefore, even when taxes are paid within the same year that they are assessed, they may incur a penalty if payments are made after July 1st. Lastly, statistical discrepancies in the data on municipal finances available from the DCED also play a part.¹⁹ On account of these reasons, in all the estimations I only use observations that lie within the 5th and 95th percentiles of the distribution for percent property taxes collected.²⁰

18 The article, http://www.philly.com/philly/news/politics/mayor/How_to_fix_Phillys_poor_tax_collection_delinquency_rate.html, for example, notes that the property tax delinquency rate for the city of Philadelphia was 6.2 percent, almost double the average for big American cities, and yet “other city tax streams have even worse compliance rates.” See also Mikesell and Liu (2011) who note that “The [delinquency] rate is slightly below 3% for most years . . .”

19 Considerable effort was taken in cleaning the data prior to use. I conducted a web search for county and municipal Comprehensive Annual Financial Reports (CAFRs) to verify the millage rates and assessed values of real estate. I consulted newspaper archives available through the ProQuest platform to look for articles referencing changes in millage, property taxes, or collection rates. Finally, I also looked through minutes of meetings for City Councils, Borough Councils, and Township Boards for any information on budgets, property taxes, and millage rates that would help corroborate or alter the data provided by the DCED. These examinations were undertaken intensively for values of percent taxes collected that fell beyond a “reasonable” range from 90–110 percent.

20 Figures A.1 and A.2 in the Online Appendix show why it is important to limit the estimation in this manner. As the figures convey, the positive relationship between tax collector compensation and collection rates is not visible when outliers are present but the positive relationship emerges when I limit the estimation to observations that lie within the 5th and 95th percentiles of the distribution of the dependent variable. I examine the robustness of the positive relationship between the commission rate and the percent property taxes collected to a variety of different approaches such as limiting the observations to those that lie within the 1st and the 99th percentiles of the distribution of the dependent variable or capping the percentage taxes collected at 110% (keeping those observations). The results obtained for the Pennsylvania Manual are included in Appendix Table A.1 and they support the conclusions of this paper. Results for the PEL sample are similar and they are available on request.

4.2 Empirical Specification

There are two analyses undertaken to examine the research questions of interest and I lay out the specification used for each. To examine if there is a relationship between tax collector compensation and performance, I look at the percentage of predicted property taxes that were actually collected by the municipality. As there is considerable year-to-year variation, I consider the average over a multiyear window. For example, when using the compensation data from the Pennsylvania Manual, I construct an average of the percent property taxes collected for the subsequent four-year period. Moreover, given that I include municipalities that pay either on a commission or a salaried basis in the same estimation, I standardize the compensation measures – the commission rate and the log of salary paid – for the two regimes to have a mean of 0 and a standard deviation of 1. The specification used is:

$$P_{it} = \alpha + \beta_1 \times C_{i(\tau-1)} \times S_{i(\tau-1)} + \beta_2 \times S_{i(\tau-1)} + \gamma \times W_{it} + \lambda_c + \gamma_\tau + \varepsilon_{it} \quad (4.1)$$

In the equation above, P_{it} is the percentage of property taxes actually collected in municipality i over a 4-year window, $C_{i(\tau-1)}$ refers to standardized compensation at the start of that 4-year window,²¹ $S_{i(\tau-1)}$ is an indicator variable that equals 1 when the municipality pays a flat salary and 0 when the municipality uses a commission, W_{it} includes municipal-level controls that are likely to have an independent influence on the efficiency of the property tax collection system, and ε_{it} is the error term. If there are differences in collection rates between municipalities that compensate on the basis of a commission versus those that compensate on the basis of a salary that are unrelated to tax collector compensation, those are captured by the coefficient β_2 . λ_c represents county fixed effects and γ_τ represents period fixed effects, corresponding to the vintage of the Pennsylvania Manual. As this dataset includes multiple observations for each municipality (and compensations do change over time), I can introduce municipality fixed effects too.²²

To examine whether a higher collection of property taxes is rewarded (or punished) by voters, I examine two different margins. I first examine whether

²¹ The length of a term for the office of tax collector is 4 years and per statute, compensation is set prior to the start of the term with little possibility for change during the term itself. Thus I use the subscript $(\tau - 1)$ to denote the start of the electoral term.

²² Compensation schemes can also change over time. Of the 226 municipalities that show up at least once in the Pennsylvania Manual data, 202 (or 89.4%) never change their compensation scheme, with 172(30) of them always using a salary-based (commission-based) scheme. Of the remaining 24 municipalities, 18 change once from a commission to a salary scheme, 2 change in the opposite direction, and only 4 experience more than one change.

the incumbent is more or less likely to run based on her performance in office. In case the incumbent decides to run again for office, I examine whether the collection rate during the period she was in office influences her vote share. With both dependent variables, I include the lagged vote share from the previous electoral cycle in half of the specifications, while in the remaining, I run a simpler specification that does not control for the lagged vote share. When the lagged vote share is controlled for, the specification used is as follows:

$$V_{it} = \alpha + \beta \times P_{i(\tau-1)} + \gamma \times V_{i(\tau-1)} + \varepsilon_{it} \quad (4.2)$$

In (4.2), V_{it} is the current vote share of an incumbent tax collector, $V_{i(\tau-1)}$ is her lagged vote share, $P_{i(\tau-1)}$ is the percentage of property taxes actually collected over her prior term (a 4-year window between electoral period $(\tau - 1)$ and τ), and ε_{it} is the error term.²³ This analysis is conducted using data on elections held between 2003 and 2009 for all 16 counties for which such data were available.

5 Results

5.1 Does Higher Compensation Induce the Collection of More Taxes?: Evidence from the Pennsylvania Manual Data

I first turn to data from the Pennsylvania Manual to examine the relationship between pay and performance for tax collectors. Table 4 presents the results obtained by pooling observations from all counties across all vintages of the Pennsylvania Manual.²⁴ Column (1) estimates a parsimonious specification with no controls but for municipal population. Column (2) adds controls for the millage rate and the log of market value of real estate per capita, as these variables may influence the percentage of property taxes collected. For example, a low market value of real estate per capita is likely in municipalities experiencing a challenging economic environment and it may be harder to collect property taxes in such a municipality.

23 Of the 731 decisions to run for office in the sample, about 85% involved a full 4-year term. For the remaining 15% that involved a 2-year term, those were driven by deaths or resignations. In those cases, the percentage of property taxes collected was averaged over the 2-year term of the incumbent.

24 Data from the Pennsylvania Manual pertain to the largest municipalities within the state and most of these municipalities pay their tax collectors a fixed salary. Depending on the vintage of data, anywhere between 13 and 31 municipalities in this dataset rely at least partly on a commission-based structure, but of those, I know the commission rate for fewer than a third of the municipalities (an average of 6 per year).

Table 4: Effect of higher compensation on percent property taxes collected (using the Pennsylvania Manual).

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Percent municipal property taxes collected (averaged over a 4-year period)						
Compensation (for collectors paid on a commission basis)	0.967** (0.387)	0.897** (0.393)	0.527* (0.317)	0.544* (0.305)	1.879* (0.986)	1.735* (0.980)
Compensation (for collectors paid on a salaried basis)	−0.242*** (0.0813)	−0.109 (0.0814)	−0.134 (0.102)	−0.0977 (0.107)	0.0918 (0.314)	0.0997 (0.310)
Municipality pays on a salaried basis	−0.473 (0.392)	0.0595 (0.398)	−0.240 (0.411)	0.0245 (0.410)	0.820 (1.112)	0.871 (1.109)
Log population	−0.159 (0.129)	−0.165 (0.125)	−0.263 (0.195)	−0.150 (0.188)	−4.089** (1.668)	−4.041** (1.666)
Millage rate		−0.0157*** (0.00329)		−0.0193*** (0.00347)		−0.0125*** (0.00472)
Log of market value of real estate per capita		0.633*** (0.147)		0.391* (0.209)		−0.406 (0.451)
Constant	101.2*** (1.310)	95.10*** (1.911)	102.0*** (1.908)	97.37*** (2.800)	138.6*** (16.25)	142.3*** (16.17)
Fixed effects	None	None	County	County	Municipality	Municipality
Number of fixed effects introduced	N/A	N/A	39	39	199	199
N	716	716	716	716	716	716
R ²	0.041	0.10	0.14	0.19	0.50	0.51

Table 4: (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>p</i> -value that effects of higher compensation differ across collectors paid on a commission basis vs. those paid on a salaried basis	0.002	0.012	0.048	0.048	0.078	0.105

Notes: Data on compensation of municipal tax collectors are from six issues of the Pennsylvania Manual: Volumes 107, 109, 111, 113, 115, and 117 and pertain to years 1985, 1989, 1993, 1997, 2001, and 2005. Because commission rates and salaries of tax collectors are measured differently (in percentage points and dollars respectively), in order to analyze the two regimes simultaneously, both compensation measures are standardized to have a mean of 0 and a standard deviation of 1. Data on percentage of predicted property taxes actually collected are based on author's calculations and are the averages for the subsequent 4-year periods (e.g. 1986–1989 for the 1985 compensation data). Period fixed effects (1986–1989, 1990–1993, etc.) are included in all specifications of the table but have not been reported. To reduce the influence of outliers, only those observations for which percent taxes collected lies within the 5th and 95th percentiles of the underlying distribution have been used. Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results in Table 4 suggest a positive and statistically significant relationship between the compensation received by tax collectors and the percentage of property taxes collected for municipalities that adopt a commission-based scheme, with a 1 percent increase in the commission rate associated with an increase in the percent taxes collected by anywhere between 0.5 and 1.9 percent depending on the specification and averaging about 1.1 percent.²⁵ In contrast, when considering the effect of a higher salary for municipalities that compensate using a flat salary, I do not find a statistically significant relationship between compensation and the percentage of property taxes collected suggesting that there is no relationship between pay and performance for these collectors. This is true regardless of whether I pool across all municipalities (cols. (1) and (2)) or introduce county fixed effects (cols. (3) and (4)) or municipal fixed effects (cols. (5) and (6)). The fact that I do not find a positive relationship when I introduce municipal fixed effects rules out the possibility that time-invariant differences in municipal characteristics are what drive the lack of a relationship between pay and performance for salaried collectors. Moreover, as the coefficients in the last row suggest, the effects of higher compensation for the commission regime are statistically distinguishable from the effects of compensation for the salary regime.²⁶

Because the estimation in Table 4 uses data from multiple issues of the Pennsylvania Manual spread over two decades, I can also look at each cross-section of the data and can confirm that the lack of a relationship between salary and performance holds for each cross-section. Those additional results are available on request.

5.2 Does Higher Compensation Induce the Collection of More Taxes?: Evidence from the PEL Data

²⁵ As noted earlier, to compare the commission and salary schemes in a single regression, the commission rate (and the log of salary) were standardized. For the observations used in the estimation, the standard deviation of the commission rate equals 0.984. Accordingly to quantify the magnitude of an increase in the commission rate by 1 percent, the coefficients in the first row of the table need to be divided by the standard deviation of 0.984.

²⁶ I also consider an additional specification as a robustness check in which I introduce both the commission rate and the log of salary for the commission scheme, whereas I only introduce the log of salary for the salary scheme. If the claim that the strength of incentives in the commission scheme is better proxied by the commission rate is correct, then one would expect the coefficient on the commission rate to be positive and statistically significant but the coefficient on the log of salary to be statistically insignificant. That is exactly what I find. Those additional results are available on request.

■Q6 Table 5 presents the results of analyzing the PEL data using specification (4.1). As before, the first (second) row provides the estimated effects of higher compensation on percent real estate taxes collected for tax collectors compensated on a commission (salary) basis. Columns (1) and (2) are identical to those in Table 4. Additionally, because of the availability of data, in Column (3) I introduce two new controls, the first being the percentage of housing units that have a mortgage. This variable is introduced because property tax payments for households with a mortgage are typically folded into a monthly payment alongside the mortgage principal, interest, and homeowners' insurance and Hayashi (2014) suggests that forcing homeowners to save a portion of their income each month for property tax payments makes homeowners less likely to become delinquent on these payments.²⁷ I also control for the local unemployment rate as a proxy for local economic conditions because it seems plausible that collection of the property tax will be lower in municipalities experiencing fiscal distress. Columns (4) through (6) replicate the pattern in columns (1) through (3) but add in county fixed effects.

As before, I continue to find a positive and statistically significant coefficient between pay and performance for collectors who are compensated on a commission basis. This is true across all six columns of the table as progressively more controls are added. In terms of magnitude, a 1 percent increase in the commission rate is associated with an increase in the percent taxes collected by anywhere between 0.5 and 0.8 percent depending on the specification.²⁸ While the relationship between pay and performance is also positive for collectors who are paid a flat salary, the relationship fails to achieve statistical significance at conventional levels of significance. However, in this instance, as the coefficients in the last row suggest, I cannot rule out the possibility that the effects of higher compensation are identical across the two regimes unlike what I obtained when analyzing the Pennsylvania Manual sample.

²⁷ Beyond controlling for the percent of homes with a mortgage, I also introduced a variable capturing the usage of escrow. The results are robust to the inclusion of this variable and are available on request. I am grateful to the Lincoln Institute of Land Policy for making these data available through specially commissioned tabulations of the 1990 and the 2000 Decennial Censuses.

²⁸ Because the commission rate was standardized, to estimate the magnitude of an increase in the commission rate by 1 percent, the coefficients in the first row need to be divided by the standard deviation.

Table 5: Effect of higher compensation on percent property taxes collected (using the PEL survey).

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Percentage municipal property taxes collected (average for years between 2005 and 2008)						
Compensation (for collectors paid on a commission basis)	0.391** (0.193)	0.448** (0.199)	0.436** (0.204)	0.603*** (0.200)	0.581*** (0.218)	0.580*** (0.222)
Compensation (for collectors paid on a salaried basis)	0.282 (0.435)	0.293 (0.424)	0.267 (0.428)	0.611 (0.439)	0.545 (0.432)	0.511 (0.433)
Municipality pays on a salaried basis	-0.356 (0.413)	-0.436 (0.430)	-0.388 (0.432)	-0.638 (0.419)	-0.571 (0.426)	-0.550 (0.420)
Log population	0.193 (0.175)	0.176 (0.178)	0.159 (0.186)	0.135 (0.198)	0.131 (0.198)	0.138 (0.201)
Millage rate		0.0144 (0.0206)	0.0185 (0.0212)		-0.0125 (0.0273)	-0.0119 (0.0277)
Log of market value of real estate per capita		0.349 (0.361)	0.312 (0.365)		0.219 (0.369)	0.241 (0.380)
Percentage of housing units with a mortgage			0.0181 (0.0188)			0.00591 (0.0192)
Local unemployment rate			0.0366 (0.0320)			0.0335 (0.0303)
Constant	98.49*** (1.396)	94.65*** (4.173)	93.88*** (4.330)	98.93*** (1.554)	96.61*** (4.623)	95.80*** (4.870)
County fixed effects	No N/A	No N/A	No N/A	Yes 8	Yes 8	Yes 8
Number of fixed effects introduced						
N	228	228	228	228	228	228
R ²	0.022	0.027	0.034	0.11	0.12	0.12

Table 5: (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>p</i> -value that effects of higher compensation differ across collectors paid on a commission basis vs. those paid on a salaried basis	0.827	0.754	0.740	0.987	0.943	0.896

Notes: Data on compensation paid to municipal tax collectors are based on a survey conducted by the Pennsylvania Economy League in 2004 in nine counties in Central and Eastern Pennsylvania. Because commission rates and salaries of tax collectors are measured differently (in percentage points and dollars respectively), in order to analyze the two regimes simultaneously, both compensation measures are standardized to have a mean of 0 and a standard deviation of 1. Data on percentage of predicted property taxes actually collected are based on author's calculations and are the averages for years 2005–2008. To reduce the influence of outliers, only those observations for which percent taxes collected lies within the 5th and 95th percentiles of the underlying distribution have been used. Robust standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Robustness Checks for Results on the Pay-Performance Relationship

In this section, I present a number of robustness checks for the result that there is a positive relationship between pay and performance for tax collectors paid on a commission basis, while no such relationship exists for tax collectors paid a flat salary.

5.3.1 Distinguishing Between Appointed and Elected Tax Collectors

Although the vast majority of tax collectors are elected officials, in a small proportion of municipalities, I find tax collectors who were appointed to their office. The most common reason for why one might find an appointed tax collector is because municipalities can adopt a home rule charter and several such charters have done away with the elected office of the tax collector, instead appointing a civil servant to collect property taxes for their municipality.²⁹ Given that the literature finds that elected and appointed officials behave differently (e.g. Besley and Coate 2003 and Whalley 2013), an interesting angle to explore using these data is whether elected and appointed tax collectors respond differently to incentives. This analysis is however impossible with the PEL survey as it excluded appointed tax collectors by design. Table 6 presents these results obtained by analyzing the Pennsylvania Manual data. Splitting by appointed versus elected status illustrates that the lack of a relationship between pay and performance for salaried tax collectors holds separately for both sets of tax collectors.

5.3.2 Using the Annual Series and Also Examining for Political Budget Cycles

Rather than using one observation per four-year period (the typical term in office) and the average percentage taxes collected during that period, in this robustness check, I exploit the fact that data on municipal finances are available on an annual basis from the Pennsylvania DCED. Using the annual nature of these data, I re-examine the relationship between tax collector compensation and the percentage of taxes collected in each year of the sample period. The results indicate that my conclusions are not sensitive to whether I use four-year averages of the percentage taxes collected or their annual values. Using the annual nature of the data also lets me examine for the presence of any “political budget cycles” or whether tax

²⁹ “A municipality which has adopted a home rule charter may exercise any powers and perform any function not denied by the Constitution of Pennsylvania, by statute or by its home rule charter.” (<http://www.legis.state.pa.us/WU01/LI/LI/CT/HTM/53/00.029..HTM>).

Table 6: Effect of higher compensation on percent property taxes collected for collectors compensated on basis of a salary (using the Pennsylvania Manual) – distinguishing between elected and appointed status.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Percentage municipal property taxes collected (averaged over a 4-year period)						
Compensation (for elected tax collectors)	-0.242*** (0.0805)	-0.127 (0.0820)	-0.151 (0.102)	-0.118 (0.107)	0.0204 (0.308)	0.0253 (0.305)
Compensation (for appointed tax collectors)	-0.855 (0.678)	-0.797 (0.670)	-0.786 (0.793)	-0.966 (0.786)	0.306 (1.025)	0.323 (0.979)
Municipality has an appointed tax collector	0.721 (0.657)	1.121* (0.625)	0.683 (0.684)	0.971 (0.657)	1.283* (0.749)	1.314* (0.733)
Log population	-0.152 (0.145)	-0.186 (0.140)	-0.259 (0.201)	-0.141 (0.193)	-4.742*** (1.800)	-4.815*** (1.798)
Millage rate		-0.0163*** (0.00330)		-0.0191*** (0.00354)		-0.0130** (0.00510)
Log of market value of real estate per capita		0.662*** (0.147)		0.439** (0.214)		-0.340 (0.440)
Constant	100.7*** (1.476)	95.10*** (1.928)	101.7*** (2.016)	96.84*** (2.819)	145.8*** (17.66)	150.1*** (17.44)
Fixed effects	None	None	County	County	Municipality	Municipality
Number of fixed effects introduced	N/A	N/A	39	39	185	185
N	682	682	682	682	682	682
R ²	0.034	0.10	0.14	0.18	0.50	0.51

Table 6: (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
<i>p</i> -value that effects of higher compensation differ across elected vs. appointed collectors	0.371	0.320	0.428	0.285	0.781	0.762

Notes: Data on salaries of municipal tax collectors are from the Pennsylvania Manual and pertain to years 1985, 1989, 1993, 1997, 2001, and 2005 and they have been standardized so as to have a mean of 0 and a standard deviation of 1. Data on percentage of predicted property taxes actually collected are based on author's calculations and are the averages for the subsequent 4-year periods. Period fixed effects are included in all specifications of the table but have not been reported. To reduce the influence of outliers, only those observations for which percent taxes collected lies within the 5th and 95th percentiles of the underlying distribution have been used. Of the 682 observations used in this estimation, 628 correspond to elected tax collectors with the remaining 54 corresponding to appointed tax collectors. All but seven of these 54 cases pertain to municipalities that have a form of home rule and as part of their Home Rule Charter, voters in those municipalities have made the decision to appoint (as opposed to elect) tax collectors. Robust standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

collection rates vary in any systematic manner over the term of a tax collector in office but there is no evidence suggesting such cycles in tax collection in either dataset.

5.3.3 Distinguishing between City Treasurers and Tax Collectors for Municipalities of all Other Classes

As noted earlier, the office of city treasurers appears to be qualitatively different from the office of treasurers in townships of the first class or tax collectors in boroughs and townships of the second class. City treasurers are eligible for health insurance, pension benefits, and other fringe benefits from their employers and therefore the salary they receive reflects only a portion of their total compensation. Hence, I repeat the entire analysis reported in Table 4 but distinguish cities from other municipal classes. The conclusion that there is no evidence of a positive relationship between salary and performance holds in both sub-samples – the first comprised cities and the second comprised boroughs and townships.

5.3.4 Limiting the Estimation to Municipalities which are Similar

As the discussion regarding the institutional set-up (and the summary in Table 1) make evident, larger municipalities tend to pay their collectors on the basis of a flat salary whereas smaller municipalities tend to pay their collectors on a commission basis. This may be a sensible approach given that even a commission rate of 1 percent in a large municipality may translate to a substantial amount – more than what is required for attracting a qualified individual to the position.³⁰ Thus, the primary distinction between municipalities which pay on a commission basis versus those that pay collectors a salary is their size. This is not a problem for us per se because I control for population in all specifications throughout the paper. Nevertheless, in this robustness check, I re-estimate the regressions over the population ranges which are common to both the commission and the salary schemes. Effectively this implies dropping the smallest municipalities (which do not compensate on the basis of a salary in my sample) as well as the largest municipalities (which do not compensate on the basis of a commission). The results I obtain by limiting myself to such municipalities are similar to what I obtain earlier; I confirm that higher compensation is associated with higher

30 A 1 percent commission rate would lie below the 25th percentile commission rate, which stands at 3.42 and 2.00 in the two datasets respectively.

collection rates, but only for municipalities that pay on a commission basis and not for municipalities that pay a flat salary.

5.3.5 Excluding Municipalities that Bunch at Caps

As described in Section 2, for townships of the first class, annual compensation for township duties alone cannot exceed \$10,000, whereas compensation in boroughs and townships of the second class can be in the form of a salary or a commission, with total compensation not to exceed 5 percent of the amount of taxes. Just over a third of the observations used in the Pennsylvania Manual sample come from townships of the first class and of those observations, about a third set their salary exactly equal to \$10,000.³¹ Likewise, in the PEL sample – which is more representative of the overall state and includes a greater proportion of municipalities that compensate their tax collectors on a commission basis – about half of the overall sample comprises townships of the second class and of those 40 percent pay their tax collectors a commission of exactly 5 percent.³²

In a robustness check, I discard all observations that bunch either at the \$10,000 salary cap or the 5 percent cap on commission rate set by statute and continue to find evidence of a positive relationship between compensation and performance for tax collectors paid on a commission basis, with no relationship for collectors paid a flat salary. To be conservative, I provide another robustness check in which I discard all observations where (1) tax collector salary equals or exceeds \$10,000 or (2) the commission rate equals or exceeds 5 percent. I continue to find similar results as those reported above.

All of the results referenced in these Sub-sections 5.3.2, 5.3.3, 5.3.4, and 5.3.5 are available in the Online Appendix. Additionally I examine whether the effects of compensation on performance are non-linear but find little evidence to suggest as much. For the range of compensations I observe in the data, the estimates continue to point to a positive (null) relationship between the commission rate (salary) and the percent of property taxes collected.³³

31 Of the 716 observations from the Pennsylvania Manual used in the estimation in Table 4, 240 (or 33.5 percent) are for townships of the first class, and of those 74 (or 30.8 percent) pay their tax collectors a salary of \$10,000.

32 Of the 228 observations used in the estimation in Table 5, 115 (or 50.4 percent) are for townships of the second class, and of those 46 (or 40.0 percent) pay their tax collectors a commission of 5 percent. In the PEL sample, only 2 of the 47 municipalities that pay a salary (or 4.26 percent) are at the cap of \$10,000, with the median salary being \$4064.

33 I also explore an instrumental variables (IV) approach to address the concern that the compensation scheme chosen may be endogenous to the property tax collection rate and other

6 Ruling out Alternative Explanations

The results in the earlier section suggest that there is a positive and statistically significant relationship between pay and performance for tax collectors who are compensated on the basis of a commission, whereas no such relationship exists for tax collectors who are paid a flat salary. In this section I consider a number of alternative possible explanations but rule them out on the basis of the available data.

6.1 Does the Relationship Run from Performance to Pay rather than in the Other Direction?

The first alternative explanation I consider is that although there is evidence of positive relationship between the commissions paid to tax collectors and the percentage of taxes they collect, perhaps this relationship operates in the other direction. In other words, it is higher performance that results in higher commission rates, rather than higher commission rates inducing higher performance.

In order to examine this hypothesis, I need panel data which therefore leads me to the Pennsylvania Manual sample. Few municipalities in this dataset pay on a commission basis (and I know the commission rate for fewer than a third of those) precluding me from analyzing this hypothesis for municipalities that compensate on a commission basis. However, for two municipalities in this sample that I observe over a 16-year window, I find no change in the commission rate paid to tax collectors during this period suggesting that changes to commission rates are infrequent. I can however examine this hypothesis more formally for municipalities that compensate their collectors with a salary. As long as it is the case that the political forces that induce higher compensation for tax collectors with better collection rates apply to municipalities regardless of whether they

municipal characteristics. In the search for a possible instrument, I exploit the fact that the class to which a municipality belongs influences the choice of a compensation scheme, with cities and townships of the first class more likely to choose a salary scheme, and townships of the second class more likely to use a commission-based scheme. Additionally population and population density play a role as well, with larger and more densely populated municipalities likelier to opt for a salary-based scheme. The results obtained using the dummy variables for municipal class, log population, and log population density as instruments line up with what we observe earlier; I continue to find a positive relationship between tax collector compensation and percent property taxes collected for municipalities that pay their collectors on a commission basis but fail to find a similar relationship for collectors paid a salary. However, the level of statistical significance and effect size varies across the two datasets and the Sargan test of overidentification is often rejected calling into question the validity of the instruments.

pay a commission or a salary, examining the performance–pay relationship for municipalities compensating on the basis of a salary is informative of the performance–pay relationship for municipalities using a commission-based structure. In the regressions presented in Table 7 I consider a variety of approaches and functional forms; I examine if either the *level* or *changes* in salary can be predicted based on the *lagged level* of percent property taxes collected or based on the *lagged change* in the percent property taxes collected. I also consider salary, in log terms (in columns (1) through (4)) and in absolute terms (in columns (5) through (8)).

Regardless of the functional form or the empirical approach used, I do not find any evidence of a positive performance-pay relationship for tax collectors who are paid a flat salary as the coefficients of interest are indistinguishable from zero in all specifications. As noted above, while this does not let me directly rule out the possibility that a positive performance–pay relationship might exist for collectors compensated on a commission basis (which therefore results in the observed positive relationship between pay and performance), that appears unlikely.

6.2 Municipalities that Pay Higher Commissions are Differ from Municipalities that Pay Lower Commissions

A second alternative explanation is that municipalities that pay higher commissions differ in ways that are observable and unobservable from municipalities that pay lower commissions and it is these differences in municipal characteristics that result in the observed positive relationship between pay and performance, rather than the additional effort exerted by tax collectors as proposed in this paper. There are a number of responses that one could offer in response to that critique. One could include municipal characteristics such as the percentage of housing units with a mortgage or proxies for economic conditions to rule out the possibility that differences in these characteristics are what drive the underlying pay–performance relationship, as was done in Table 5.

Beyond these responses, I conduct a falsification exercise that helps reduce the possibility that differences in unobservable characteristics are what drive the pay–performance relationship. As was noted earlier, Pennsylvania’s municipalities rely primarily on two taxes – the property tax and the local earned income tax – with each contributing roughly 40–45 percent of all tax revenue. If municipalities that pay their tax collectors more differ in ways that are unobservable from those that pay their collectors less and these differences result in higher collection rates for the property tax, then perhaps such municipalities will also exhibit higher compliance with the earned income tax. Hence I can estimate

Table 7: Presence of a reverse causal relationship between pay and performance - analyzed for collectors compensated on basis of a salary (using the Pennsylvania Manual).

	(1)	(2)	(3)	(4)
	D.v.: Log annual salary		D.v.: Change in log annual salary	
Lagged percent property taxes collected	0.00913 (0.00788)		0.00161 (0.00450)	
Lagged change in percent property taxes collected		0.00448 (0.00503)		−0.00240 (0.00379)
Constant	8.579*** (0.797)	9.456*** (0.0443)	−0.0854 (0.454)	0.0569*** (0.0185)
N	489	379	456	345
R ²	0.78	0.86	0.41	0.50
Municipal fixed effects	Yes	Yes	Yes	Yes
Number of fixed effects introduced	146	132	150	131
	(5)	(6)	(7)	(8)
	D.v.: Annual salary		D.v.: Change in annual salary	
Lagged percent property taxes collected	84.04 (98.57)		39.78 (60.98)	
Lagged change in percent property taxes collected		39.79 (60.26)		27.49 (44.73)
Constant	8033.8 (9987.9)	15963.2*** (547.0)	−2845.2 (6168.7)	1110.1*** (283.5)
N	489	379	454	343
R ²	0.80	0.89	0.48	0.48

Table 7: (continued)

	(5)	(6)	(7)	(8)
		D.v.: Annual salary		D.v.: Change in annual salary
Municipal fixed effects	Yes	Yes	Yes	Yes
Number of fixed effects introduced	146	132	146	127

Notes: Data on salaries of municipal tax collectors are from the Pennsylvania Manual and pertain to years 1985, 1989, 1993, 1997, 2001, and 2005. Data on percentage of predicted property taxes actually collected are based on author's calculations and are averages over 4-year periods. Period fixed effects and municipal fixed effects are included in all specifications of the table but have not been reported. To reduce the influence of outliers, only those observations for which the dependent and independent variables lie within the 5th and 95th percentiles of their respective distributions have been used. Robust standard errors in parentheses $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

regressions similar to those presented earlier but now consider compliance with the earned income tax as the dependent variable of interest. If I fail to find a positive relationship between tax collector compensation and compliance with this local income tax, it reduces the likelihood that unobservable differences in municipal characteristics are what drive the pay-performance relationship.³⁴

Operationalizing this exercise necessitates obtaining information on the earned income tax rate, the size of the economic base, and earned income taxes that were actually collected. Data on the earned income tax rate and any income-based exemptions are drawn from reports prepared by Pennsylvania's Department of Community and Economic Development (DCED).³⁵ Given that this tax applies only to earned income and not to income such as dividends or interest or payments for disability benefits or unemployment compensation, I obtain the number of households that have positive earnings and the level of mean earnings for such households from the 2000 Decennial Census. Lastly, data on earned income taxes actually collected are from the annual municipal finance reports prepared by the Pennsylvania DCED. I use these data to define a measure of performance, the percentage of earned income taxes collected as:³⁶

34 The validity of this falsification exercise hinges on compensation across the different taxes not being correlated with each other. Using historical data on the costs of tax collection available from the Pennsylvania DCED, I in fact confirm that this assumption is true – there is *no* statistically significant relationship between the compensation received by tax collectors and these overall costs of tax collection, which are largely driven by the costs of collecting the earned income tax. These additional results are available on request from the author.

35 Unfortunately, historical data are not available, and the data applies to the current year (2019). However, because earned income tax rates themselves change infrequently, earned income tax rates during the period of interest (2005–2008) are likely to be correlated with the earned income tax rate in 2019.

36 This measure is likely to capture compliance with the earned income tax imperfectly. However, I am not aware of any alternative source of data on collection rates that could be used instead. Examining compliance with this tax is a useful exercise as there is significant variation in compliance for two reasons. First, the self-employed are outside the purview of withholding and they contribute to non-compliance and second, out-of-state employers are not required to withhold local income taxes for their Pennsylvania employees. Anecdotal evidence suggests that in some jurisdictions that lie near Pennsylvania's borders with other states, about 35 percent of local income taxes due to the district were not paid on time. <https://www.mcall.com/news/mc-xpm-2007-05-06-3732456-story.html> It is in response to poor collection rates of the earned income tax, that the state decided to consolidate the number of earned income tax collectors from about 560 to 70 but those changes came into effect in January 2012 – well after the period I analyze in Table 8. Notably no changes were made to the property tax collection system at the time.

Table 8: Effect of higher compensation of tax collectors on percent earned income taxes collected (identical time period as in the PEL survey).

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Percentage municipal earned income taxes collected						
(average for years between 2005 and 2008)						
For municipalities compensating on a commission basis						
Compensation	-1.880 (1.509)	-1.004 (1.409)	-1.409 (1.386)	-0.0194 (1.662)	0.592 (1.652)	-0.133 (1.673)
Log population	3.588*** (1.290)	1.724 (1.300)	1.069 (1.206)	2.973*** (1.372)	2.346* (1.299)	1.972 (1.233)
Millage rate		0.189 (0.145)	0.252* (0.150)		0.271 (0.218)	0.298 (0.225)
Log of market value of real estate per capita		19.28*** (3.661)	17.00*** (3.539)		19.53*** (4.093)	18.44*** (3.787)
Percentage of housing units with a mortgage			0.348*** (0.125)			0.332** (0.147)
Local unemployment rate			0.153 (0.297)		0.248 (0.297)	
Constant	94.11*** (10.05)	-106.1*** (38.55)	-98.69*** (37.43)	98.13*** (10.49)	-114.6** (44.69)	-121.6*** (42.30)
<i>N</i>	185	185	185	185	185	185
<i>R</i> ²	0.052	0.22	0.25	0.14	0.27	0.29

Table 8: (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
County fixed effects	No	No	No	Yes	Yes	Yes
Number of fixed effects introduced	N/A	N/A	N/A	8	8	8

Notes: Data on commission rates paid to municipal tax collectors is based on a survey conducted by the Pennsylvania Economy League in 2004 in nine counties in Central and Eastern Pennsylvania and the rates have been standardized so as to have a mean of 0 and a standard deviation of 1. Data on percentage of predicted earned income taxes actually collected are based on author's calculations and are the averages for years 2005–2008. To reduce the influence of outliers, only those observations for which percent taxes collected lies within the 5th and 95th percentiles of the underlying distribution have been used. Robust standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Percentage of earned income taxes collected

$$= \frac{\text{Earned income taxes collected}}{(\text{Tax rate}/100 \times \text{base of income tax})} \times 100$$

where, base of income tax = number of households with positive earnings $\times$ (mean earnings – income exemption).

Table 8 presents these results and they suggest that there is no relationship between the commission rate for the property tax and compliance with the earned income tax. This result is reassuring in that it reduces the likelihood that differences in unobservable characteristics across municipalities are what drive the positive relationship between commission rates and collection of property taxes I document in this paper.

7 Does Higher Percent Collection while in Office Impact the Likelihood of Being Re-Elected?

One of the robust findings across the two datasets is the lack of a statistically significant positive relationship between pay and performance for tax collectors compensated on a salaried basis. This is contrary to the notion which suggests that paying workers more reduces shirking because of the higher cost of being fired (e.g., Becker and Stigler 1974; Besley 2004; Shapiro and Stiglitz 1984). What might help reconcile these findings with the theory is the lack of electoral incentives; thus, if voters appear not to reward a higher collection of property taxes at the ballot box then higher salaries may not elicit higher performance.³⁷

In order to examine the question of whether better performance by tax collectors induces electoral rewards from voters, I turn to data on elections for the office of tax collector. I narrow my search to elections held between the years 2003 and 2009 as the performance measure – percentage of taxes collected – can be constructed for this time period and because it is difficult to obtain local election data prior to the 2000s. Drawing on data available from the various county Boards of Elections, I am able to observe 726 municipalities spread across 16 counties at

³⁷ Higher collection rates are likely to be just one measure of performance. Tax collectors provide customer service by answering questions about complex tax bills, contacting mortgage companies that have paid the wrong amount in order to prevent a taxpayer from having to pay a tax penalty, and personally going to the homes (or assisted living and nursing homes) of the elderly and disabled to receive property tax payments and percentage property taxes collected does not capture any of those dimensions. However, I do not have any data on such services that are provided by these tax collectors and accordingly cannot comment on the same.

least once over the sample period.³⁸ For the municipalities for which I have data, I observe more than one election for 714 municipalities.³⁹ In order to see if voters reward higher collection of taxes, a possible dependent variable of interest is the re-election probability for an incumbent. However, as the summary statistics in Table 3 indicate, the probability that an incumbent tax collector running for office is re-elected is very high at 0.996 and therefore I cannot meaningfully comment on how re-election probabilities respond to performance in office. In the absence of that, I turn to other proxies of electoral rewards.

I first examine the decision of an incumbent to run for office again. This is based on a literature which suggests that politicians retire strategically at higher rates when they expect a challenging race (e.g. Stone et al. 2010). Panel A of Table 9 therefore presents results from a probit which examines whether the decision of an incumbent to run for office again is influenced by her performance or not. This is defined only for those municipalities where I observe the choice of the incumbent to either run again or step down, that is for all elections but the last election in the sample for a given municipality. I examine if this decision is responsive to the percentage of property taxes collected between the two election cycles.

In Panel B of Table 9, I examine how the incumbent tax collector's vote share in an election cycle is influenced by her performance in the previous term in office for races where she faces off against an opponent in the general election. Few elections to the office of tax collector are contested in practice and hence this measure is defined only for the 42 cases where the incumbent decides to run again for office and faces an opponent. In column (1), I consider the most parsimonious specification and only include the key independent variable, the percentage of taxes collected during the last term in office. Column (2) adds in a control for the lagged vote share as it is likely that the vote share of candidates is correlated over time because of party allegiances of voters⁴⁰ and because of candidates' reputations. Columns (3) and (4) parallel columns (1) and (2) but they add in county fixed effects.

38 These counties are spread across Pennsylvania and a full list is provided as Notes to Table 3.

39 Even though the election data span a 6-year period from 2003 to 2009 and the term of office of a tax collector is 4 years, I observe elections only once for 12 of the 726 municipalities. The primary reason for this drop-off is that in some municipalities no one may have run for the office in a particular cycle and in that case the municipality would have *delegated* the task of collecting taxes to someone. Such delegations are not captured in these data.

40 Elections for tax collectors are held on party lines and vote shares tend to track those of higher-profile races.

Table 9: Effect of percent taxes collected on the probability that incumbent runs and the vote share if she runs and faces an opponent.

	(1)	(2)	(3)	(4)
Panel A: Probability that the incumbent runs again for office				
Percent taxes collected during the Previous term in office	−0.00633 (0.0304)	−0.00631 (0.0304)	0.00491 (0.0322)	0.00508 (0.0322)
Lagged vote share in previous election cycle		−0.138 (0.443)		−0.112 (0.457)
Constant	1.023 (3.066)	1.145 (3.070)	−0.126 (3.246)	−0.0431 (3.242)
<i>N</i>	646	646	646	646
Pseudo- <i>R</i> ²	0.0015	0.0017	0.020	0.020
County fixed effects	No	No	Yes	Yes
Number of fixed effects introduced	N/A	N/A	15	15
Panel B: Incumbent's vote share if she runs for office and faces an opponent				
Percent taxes collected during the Previous term in office	1.724* (0.970)	1.503 (0.925)	0.980 (1.597)	0.623 (1.540)
Lagged vote share in previous election cycle		0.124* (0.0620)		0.151 (0.0935)
Constant	−98.85 (95.87)	−87.06 (91.62)	−25.48 (159.4)	−2.412 (154.0)
<i>N</i>	40	40	40	40
<i>R</i> ²	0.075	0.12	0.22	0.27
County fixed effects	No	No	Yes	Yes
Number of fixed effects introduced	N/A	N/A	10	10

Notes: The analysis is based on data for elections held between 2003 and 2009 for 16 counties (Berks, Butler, Centre, Fulton, Lancaster, Lehigh, Luzerne, Mercer, Monroe, Montgomery, Northampton, Pike, Schuylkill, Venango, Washington, and Westmoreland) gathered from the respective County Boards of Elections. The dependent variable in Panel A is the decision of an incumbent to run for office (Yes = 1, No = 0). The dependent variable in Panel B is the vote share of an incumbent who runs for office and is faced with at least one opponent in the general election (beyond write-in candidates). The lagged vote share variable in Panel A has been rescaled by dividing the percent vote share by 100 for ease of presentation. Year of election fixed effects are included in all specifications in panels A and B of the table but have not been reported. To reduce the influence of outliers, only those values of the performance measure that lie within the 5th and 95th percentiles of the underlying distribution have been used. Robust standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Based on the results in Panel A, I observe that the decision of the incumbent to run for office again is *not* influenced by her performance in office, measured as the percentage of property taxes collected between two electoral cycles. It also

does not seem to be influenced by the percentage of votes she received in the previous election. I obtain similar results as those in Panel A when I use a linear probability model (LPM) or retain a probit specification but include municipal population as an additional control, with results in the Online Appendix.

The results in Panel B when I only consider races where there was a contest in the general election suggest that the vote share of incumbents does not respond robustly to the percent of taxes collected between two election cycles as the coefficient of interest is statistically insignificant in three of the four columns and marginally significant (at the 10% level) in one of them.⁴¹ The lagged vote share of the incumbent is positive in cols. (2) and (4) (but statistically significant only in (2)) perhaps suggesting some persistence in vote shares over successive elections.⁴²

Overall, based on the descriptive evidence regarding the incumbency advantage and the results presented in Table 9, I can conclude that voters are unlikely to reward (or punish) tax collectors for their performance in office and therefore electoral considerations are unlikely to influence decisions by the tax collector of how much effort to exert. These results may therefore offer an explanation for the earlier finding that the performance of salaried tax collectors does not go up with an increase in their salary.

8 Discussion and Conclusions

This paper presents one of the first analyses of whether a pay-performance relationship exists for elected officials in the U.S. Using data on percentage of property taxes that are actually collected, the paper suggests that municipal tax collectors who are paid on a commission basis collect more in taxes as the commission they receive goes up. In contrast, there appears to be no statistically significant relationship for tax collectors who are compensated on the basis of a flat salary. This final observation can perhaps be explained by the fact that voters do not seem to reward the higher collection of property taxes. Incumbents enjoy re-election probabilities approaching 1 and neither their likelihood of running for office nor

41 Per col. (1), an increase in the percent taxes collected by 1 percent results in an increase in the votes received by 1.7 percent, where the mean and standard deviation of the dependent variable are 67 percent and 9 percent respectively. Thus, the economic significance of an increase in percent taxes collected on the vote share appears fairly modest.

42 As cols. (1)–(4) of Panel B indicate, I use only 40 observations (and not 42). The reason for the drop-off is that given the large dispersion in percent taxes collected, some of which is due to measurement error, I limit my estimations to those observations that fall between the 5th and 95th percentiles of the underlying distribution.

their vote share appear responsive to the percentage of property taxes collected between two electoral cycles.

Notwithstanding the strengths of this paper that have to do with having high quality data on compensation for tax collectors and the ability to construct a measure of productivity, it also has a number of limitations. For instance, one might be concerned that the unobserved differences between municipalities that adopt a salary scheme versus those that use a commission are what drive the difference in effects of compensation for the two schemes. Absent a regression discontinuity style set-up, where the choice of a scheme changes discontinuously at some threshold, it is impossible to rule out these concerns entirely. A second limitation pertains to the lack of data on compensation received by tax collectors from counties and school districts on whose behalf they collect property taxes. Although as noted in footnote 17, I do not believe that the lack of these data induces bias in our estimates, the conclusions regarding the relationship between pay and performance would be more robust if we did have a more comprehensive measure of pay that included compensation from *all* local government units, including counties and school districts. A final limitation pertains to the measurement error in the dependent variable, which necessitated using only those observations for which percent taxes collected lay within the 5th and 95th percentiles of the distribution.

Commenting on the welfare implications of tax collectors collecting more in property taxes when paid in the form of a commission is tricky as it involves comparing the marginal benefit of a dollar in private hands with the marginal benefit of a dollar in the hands of the government. That, in turn, is likely to hinge on one's views of government: whether one views it as a benevolent maximizer of social welfare or as a Leviathan, à la Brennan and Buchanan (1980). Moreover, a major issue with compensating tax collectors on the basis of revenues collected is that it may distort performance on non-revenue dimensions that impact citizen satisfaction. Providing high-powered incentives for tax collection raises concerns about over-taxation, extortion, and potential loss of tax morale. Unhappiness with tax farming contributed to the French revolution (White 2004) and concerns over tax-farming caused a move away from high-powered incentives towards flat compensation for tax officials (Parrillo 2013). Therefore examining how elected tax collectors in Pennsylvania internalize both the economic and political dimensions of their performance and strike a balance between revenue gains and political costs is an important question, but one left for future research, as are questions of performance along non-revenue dimensions such as customer service.

A final question to consider as I conclude the paper is whether the institution of municipal tax collectors analyzed here is so unique that the conclusions drawn have limited generalizability about the behavior of elected officials. I do not

- Q4 believe that such is the case. As Zelle (2019) notes “Part-time government is the rule, not the exception, for cities in the United States” and this is true more broadly outside the United States as well.⁴³ Likewise, although the compensation levels I observe here appear low, they are not unusual; leveraging data collected by the National Conference of State Legislatures, I know that although a few big states have full-time legislatures with higher pay, in most states, legislators are paid as if it were a part-time job with 30 states paying \$30,000 a year or less to their legislators.⁴⁴ Compensation within local government tends to be even lower with paid elected officials typically earning \$5000 to \$20,000 a year in many small towns.^{45,46} Thus, both when it comes to the level of compensation as well as their part-time nature, the municipal tax collectors that I analyze in this paper resemble elected officials found in a variety of other contexts.

In closing, the work described in the paper sheds some light on the question of whether and how incentives matter in the public sector. We would benefit from examining other such settings where we can construct credible measures of the performance of public officials, especially ones where compliance rates are lower than what we observe with the property tax.

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43 As Ferraz and Finan (2011) note “In most [Brazilian] municipalities, being a legislator is not full-time work. . . Interestingly, this pattern of dual employment is common in many other countries such as Italy, France, Germany, and the UK, where even at the federal level elected members of parliament continue to work in the private sector.”

44 <https://www.npr.org/2017/01/09/508237086/low-pay-in-state-legislatures-means-some-cant-afford-the-job>.

45 <https://www.kiplinger.com/article/business/T043-C000-S002-how-to-run-for-local-office.html>.

46 Within Pennsylvania, the median salary for the highest elected official in cities of the third class, boroughs, townships of the first class, and townships of the second class that are a part of the Pennsylvania Manual sample is \$3600; \$3000; \$3250; and \$3250, respectively (Source: Author calculations based on the Pennsylvania Manual).

the University of Michigan, the American Economic Association Annual Meeting, the International Institute of Public Finance Annual Congress, the National Tax Association Conference, and the Society of Labor Economists Annual Meeting for comments. All links were current as of May 6th 2021. Results from additional specifications described in the text but not included in the paper are available in the Online Appendix.

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Supplementary Material: The online version of this article offers supplementary material (<https://doi.org/10.1515/bejeap-2020-0008>).